

JEE MAIN CRASH COURSE

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- ✓ **CONCEPT VIDEOS**
- ✓ **LIVE DOUBT SESSIONS**
- ✓ **DPP, REVISION SHEETS**
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Designed by
Math Maestro

Anup Sir



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- ✓ CONCEPT BUILDING VIDEOS
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- ✓ 2 LEVELS OF SHEETS,
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JEE Mains 2020 Jan Chapter wise Question Bank

Three Dimensional Geometry

Q1

Find image of point (2, 1, 6) in the plane containing points (2, 1, 0), (6, 3, 3) and (5, 2, 2)

बिन्दुओं (2, 1, 0), (6, 3, 3) तथा (5, 2, 2) को समाहित करने वाले समतल में बिन्दु (2, 1, 6) का प्रतिबिम्ब है—

- (1) (6, 5, -2) (2) (6, -5, 2) (3) (2, -3, 4) (4) (2, -5, 6)

7th Jan Morning

Sol

(1)

Plane is समतल $x + y - 2z = 3 \Rightarrow \frac{x-2}{1} = \frac{y-1}{1} = \frac{z-6}{-2} = \frac{-2(2+1-12-3)}{6} \Rightarrow (x, y, z) = (6, 5, -2)$

Q2

If $Q\left(\frac{5}{3}, \frac{7}{3}, \frac{17}{3}\right)$ is foot of perpendicular drawn from $P(1, 0, 3)$ on a line L and if line L is passing through $(\alpha, 7, 1)$, then value of α is

7th Jan Evening

Sol

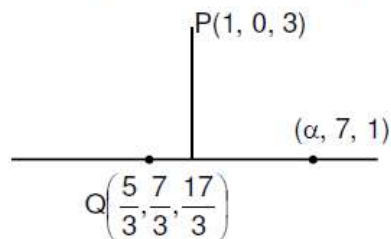
4

Three Dimensional Geometry

JEE Mains 2020 Jan Chapter wise Question Bank

Since PQ is perpendicular to L, therefore

चूँकि PQ रेखा L के लम्बवत् है इसलिए



$$\left(1 - \frac{5}{3}\right)\left(\alpha - \frac{5}{3}\right) + \left(-\frac{7}{3}\right)\left(7 - \frac{7}{3}\right) + \left(3 - \frac{17}{3}\right)\left(1 - \frac{17}{3}\right) = 0$$

$$\Rightarrow \frac{-2\alpha}{3} + \frac{10}{9} - \frac{98}{9} + \frac{112}{9} = 0$$

$$\Rightarrow \frac{2\alpha}{3} = \frac{24}{9} \Rightarrow \alpha = 4$$

03

Shortest distance between the lines $\frac{x-3}{1} = \frac{y-8}{4} = \frac{z-3}{22}$, $\frac{x+3}{1} = \frac{y+7}{1} = \frac{z-6}{7}$ is

- (1) $3\sqrt{30}$ (2) $2\sqrt{30}$ (3) $\sqrt{30}$ (4) $4\sqrt{30}$

रेखाओं $\frac{x-3}{1} = \frac{y-8}{4} = \frac{z-3}{22}$, $\frac{x+3}{1} = \frac{y+7}{1} = \frac{z-6}{7}$ के मध्य की दूरी है—

- (1) $3\sqrt{30}$ (2) $2\sqrt{30}$ (3) $\sqrt{30}$ (4) $4\sqrt{30}$

8th Jan Morning

Sol

(1)

$$\vec{AB} = 6\hat{i} + 15\hat{j} + 3\hat{k}$$

$$\vec{p} = \hat{i} + 4\hat{j} + 22\hat{k}$$

$$\vec{q} = \hat{i} + \hat{j} + 7\hat{k}$$

$$\vec{p} \times \vec{q} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 4 & 22 \\ 1 & 1 & 7 \end{vmatrix} = 6\hat{i} + 15\hat{j} - 3\hat{k}$$

$$\text{S.D.} = \frac{|\vec{AB} \cdot (\vec{p} \times \vec{q})|}{|\vec{p} \times \vec{q}|} = \frac{|36 + 225 + 9|}{\sqrt{36 + 225 + 9}} = 3\sqrt{30}$$

04

Image of $(1, 2, 3)$ w.r.t a plane is $\left(\frac{-7}{3}, \frac{-4}{3}, \frac{-1}{3}\right)$ then which of the following points lie on the plane

$(1, 2, 3)$ का किसी समतल में प्रतिबिम्ब $\left(\frac{-7}{3}, \frac{-4}{3}, \frac{-1}{3}\right)$ तो निम्न में से कौनसा बिन्दु समतल पर स्थित नहीं है :

(1) $(-1, 1, -1)$

(2) $(-1, -1, -1)$

(3) $(-1, -1, 1)$

(4) $(1, 1, -1)$

8th Jan Evening

Sol

(4)

d.r of normal to the plane

समतल के लम्बवत् द्विकअनुपात

$$\frac{10}{3}, \frac{10}{3}, \frac{10}{3}$$

$$1, 1, 1$$

midpoint of P and Q is $\left(\frac{-2}{3}, \frac{1}{3}, \frac{4}{3}\right)$

P तथा Q का मध्य बिन्दु $\left(\frac{-2}{3}, \frac{1}{3}, \frac{4}{3}\right)$

equation of plane $x + y + z = 1$

समतल का समीकरण $x + y + z = 1$

05

If points A $(2, 4, 0)$, B $(3, 1, 8)$, C $(3, 1, -3)$, D $(7, -3, 4)$ are four points then projection of line segment AB on line CD.

9th Jan Morning

Sol

8

$$\overrightarrow{AB} = (\hat{i}) - (3\hat{j}) + 8\hat{k}$$

$$\overrightarrow{CD} = 4\hat{i} - 4\hat{j} + 7\hat{k}$$

$$(\overrightarrow{AB} \cdot \overrightarrow{CD}) = \frac{4 + 12 + 56}{\sqrt{16 + 16 + 49}} = \frac{72}{9} = 8$$

06

Let the distance between plane passing through lines $\frac{x+1}{2} = \frac{y-3}{2} = \frac{z+1}{8} = 8$ and $\frac{x+3}{2} =$

$\frac{y+2}{1} = \frac{z-1}{\lambda}$ and plane $23x - 10y - 2z + 48 = 0$ is $\frac{k}{\sqrt{633}}$ then k is equal to

माना सरल रेखाओं $\frac{x+1}{2} = \frac{y-3}{2} = \frac{z+1}{8} = 8$ तथा $\frac{x+3}{2} = \frac{y+2}{1} = \frac{z-1}{\lambda}$ से गुजरने वाले समतल तथा समतल

$23x - 10y - 2z + 48 = 0$ के मध्य दूरी $\frac{k}{\sqrt{633}}$ है तब k का मान है—

(1) 1

(2) 2

(3) 3

(4) 4

9th Jan Evening

Sol

(3)

Lines must be intersecting

$$\Rightarrow (2s - 1, 3s + 3, 8s - 1) = (2t - 3, t - 2, \lambda t + 1)$$

$$2s - 1 = 2t - 3, 3s + 3 = t - 2, 8s - 1 = \lambda t + 1 \Rightarrow t = -1, s = -2, \lambda = 18$$

distance of plane contains given lines from given plane is same as distance between point $(-3, -2, 1)$ from given plane.

$$\text{Required distance equal to } \frac{|-69 + 20 - 2 + 48|}{\sqrt{529 + 100 + 4}} = \frac{3}{\sqrt{633}} = \frac{k}{\sqrt{633}} \Rightarrow k = 3$$

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